Colorado State University

Vehicle Performance Requirements Setting

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Overview

- Requirements for Vehicle Performance
- Vehicle Performance Modeling
  - Acceleration
  - Gradability
  - Fuel Economy
  - Many More
- Takeaway/Conclusions

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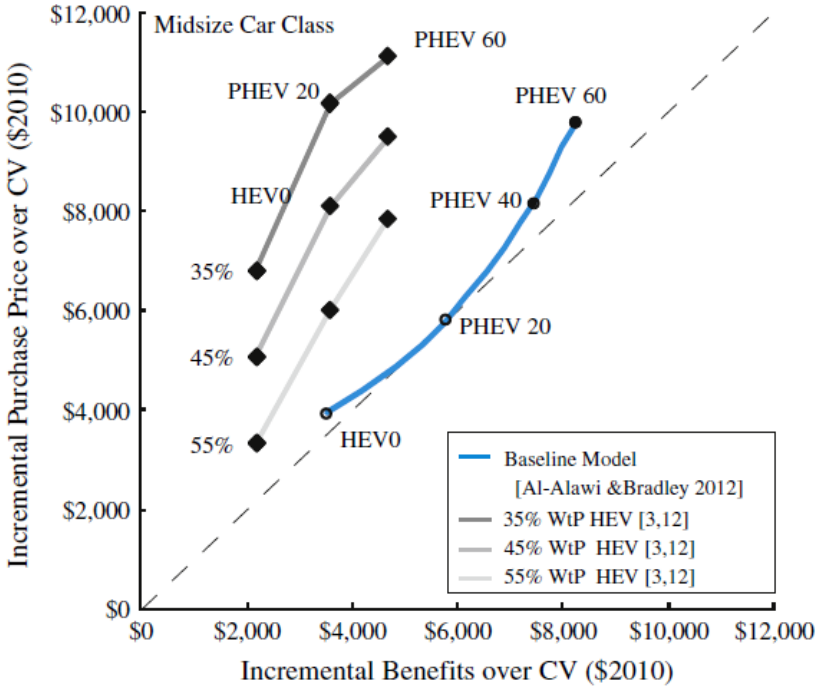
Requirements for Vehicle Performance

- The automotive industry is first and foremost a consumer products industry
  - If you cannot sell your vehicle, you will not be in business long
- Automakers derive their product requirements from
  - Consumers studies
  - Survey
  - Comparison to competition



http://www.engr.colostate.edu/~thb/Publications/tco.pdf

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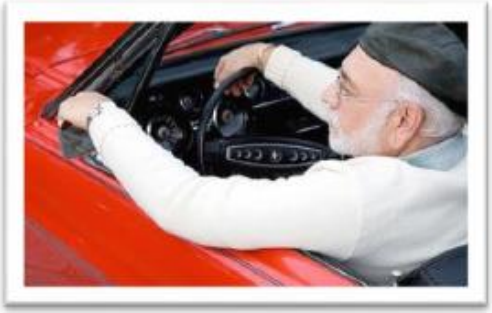
Who Is The Sports Car Driver?

CHEVROLET

Defiantly Proud- *"I am who I am, take it or leave it"*

Their cars make as much of a statement about himself or herself as a tattoo or piercing, a la "here I am." They don't want a family sedan because their vehicle also serves as an escape from the status quo of the daily grind and everyday routine (work, family, commitments).



COUPE	SEDAN
Emotional	Practical
Speed	Comfort
Sporty	Luxury
Fun	Boring
Sexy	Ordinary
Youthful	Mature
Single	Family

Source: 2014 TMFC Research

Muscle car/enthusiast psychographics

OPTIMISTIC

SELF-PROCLAIMED RISK TAKERS

CONFIDENT

DO IT, BUT DO IT DIFFERENTLY

LITTLE BIT OF BAD ASS

LIFE IS AN ADVENTURE

NIXON

MARC ECKŌ CUT & SEW

OAKLEY

BELGIAN WHITE BELGIAN STYLE WHEAT ALE BLUE MOON BREWING CO.

Ketel One VODKA

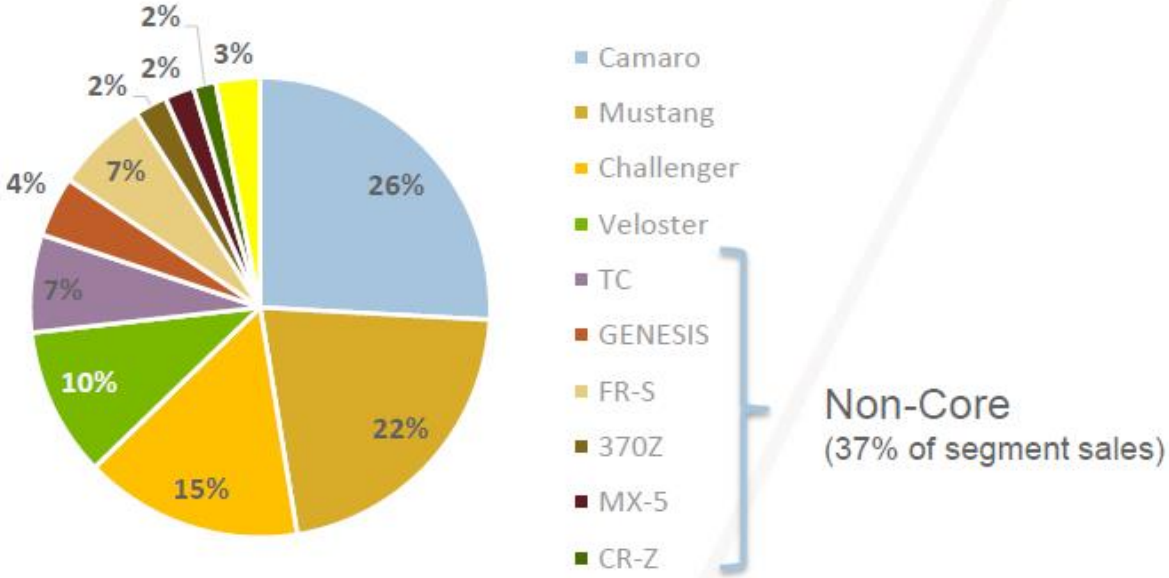
QUIKSILVER

THE STORY OF THE BRAND MATTERS

NON-CORE Opportunities

To determine opportunity with non-domestic sports cars, we looked at the group of "non-core" imported sports cars:

2013 SPORT SEGMENT RETAIL REGISTRATIONS



NON-CORE Needs/Priorities Differ

Non-core vehicle owners still value fun to drive and styling, but they also care more about fuel economy and value for the money than core segment competitors.

Reasons for Purchase

CAMARO	MUSTANG	CHALLENGER	NON-CORE
Quality of Vehicle, Vehicle Reputation and Exterior Styling	Prestige, Power and Acceleration and Vehicle Reputation	Engine Performance, Experience with Model and Exterior Styling	Fuel Economy, Reliability and Value for the Money

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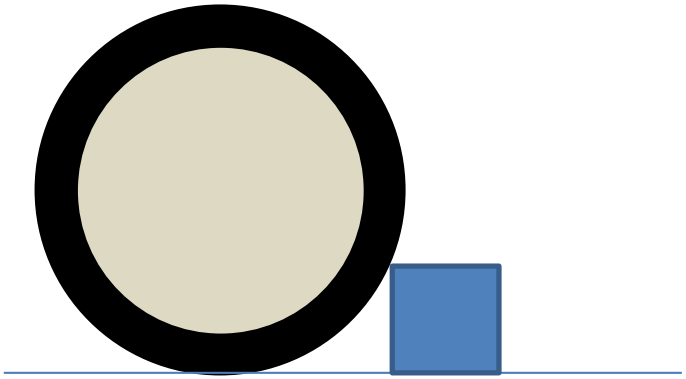
Requirements for Vehicle Performance

Metrics of comparison

- Acceleration (0-60mph, 50-70mph) units of sec
- Gradability
  - Ability to curb climb (units of cm)
  - Starting Gradability
  - Highway speed grade climbing (units of mph at X% grade)



Eisenhower Tunnels	I-70	11,158’	6% W
Approaches			
Rabbit Ears	U.S. 40	9,426’	6.8% N

- Towing Performance
  - Trailer Tow Gradability (units of units of mph at X% grade at rated capacity)
  - Starting gradability (units ofX% grade at rated capacity)

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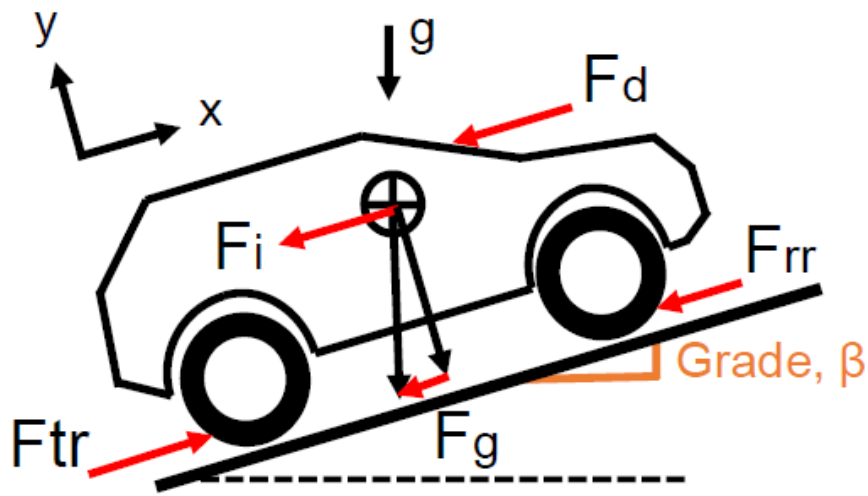

Figure 2 Free body diagram of a vehicle.

$$F_{tr} = F_i + F_{rr} + F_d + F_g$$ Equation 4

$$F_i = m * a * k_i$$ Equation 5

$$F_{rr} = m * g * (C_{rr0} + C_{rr1} * v)$$ Equation 6

$$F_d = \frac{A_f * C_d * \rho * v^2}{2}$$ Equation 7

$$F_g = m * g * \beta$$ Equation 8

Where F_{tr} = supplied tractive force,
F_i = inertial force,
F_{rr} = rolling resistance force,
F_d = drag force,
F_g = grade force,
m = vehicle mass,
a = vehicle acceleration,
k_i = inertial mass factor,
g = acceleration due to gravity,
C_{rr0} = static constant of rolling resistance,
C_{rr1} = linear constant of rolling resistance,
A_f = vehicle frontal area,
C_d = vehicle drag coefficient
ρ = density of air,
v = vehicle velocity,
β = roadway grade.

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Requirements for Vehicle Performance

Metrics of comparison could include

charging time (mins)

Fuel Economy

&

Many others that we wont really evaluate in this class: driveability, handling, temperature constraints, safety

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Measuring Fuel Economy

- Fuel Consumption [gals/mile, L/100km] = (Fuel Economy [miles/gal])$^{-1}$
- There are standards that exist for the way to measure vehicle fuel economy codified by the US Environmental Protection Agency (EPA), and Society for Automotive Engineers (SAE)

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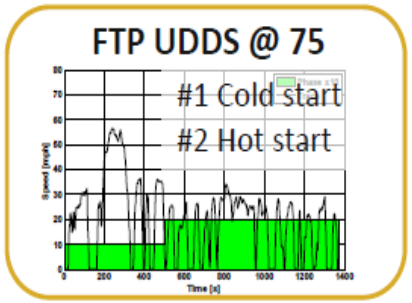
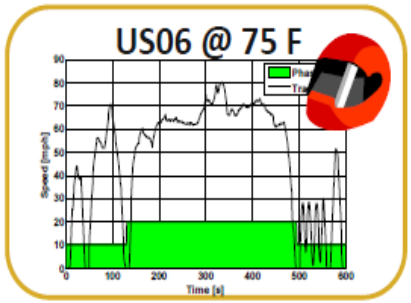
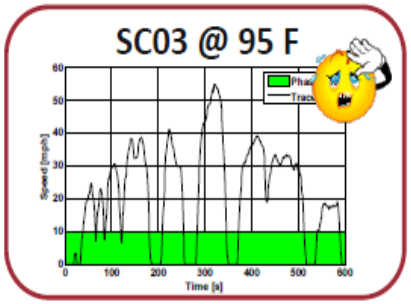
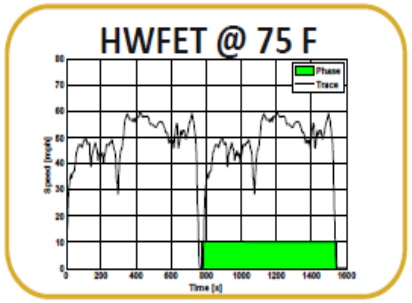
Measuring Fuel Economy

- Typical Fuel Consumption is measured in EPA 5-cycle tests
  - Are composed of a weighted combination of 5 cycles (UDDS, HWFET, SC03, US06)

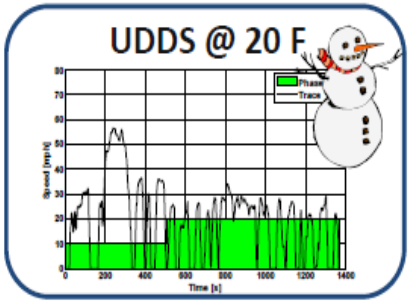








All 5 cycles existed previously for emissions testing purposes (but had not all been used to calculate fuel economy)



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Measuring Fuel Economy

- For our purposes, modeling these temperature dependencies is out of scope.
- Instead, we will use the pre-2008 FE metrics which are based off the City and Highway Cycles (UDDS, and HWFET)
- Measure FC_{UDDS}, FC_{HW} from the two drive cycles

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Measuring Fuel Economy

- Adjusted Fuel Consumption adjusts the fuel economy from the simulation to represent more "real-world" performance

$$FC_{HWY-ADJ} = \frac{1}{0.78} * FC_{HWY}$$

$$FC_{URBAN-ADJ} = \frac{1}{0.9} * FC_{URBAN}$$

- This is good metric if you are trying to model the FC that people actually get in the real world

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Measuring Fuel Economy

- Weighted, adjusted fuel consumption gives you a "single number" to represent the fuel consumption.

$$FC_{weighted,\ adj} = 0.45 * FC_{HWY-ADJ} + 0.55 * FC_{URBAN-ADJ}$$

- It represents that you drive 55% of the time on the city cycle, and 45% of the time on the highway cycle.

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Takeaway/Conclusions

- Use your understanding of the consumer to guide you to make requirements for your vehicle.
- In this class concentrate on vehicle-level requirements for powertrain and vehicle.
- You will be able to test these requirements using modeling, simulation, testing.
  - Choose the right level of modeling to meet your goals for accuracy and time budget.

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